

QMST5336: ANALYTICS

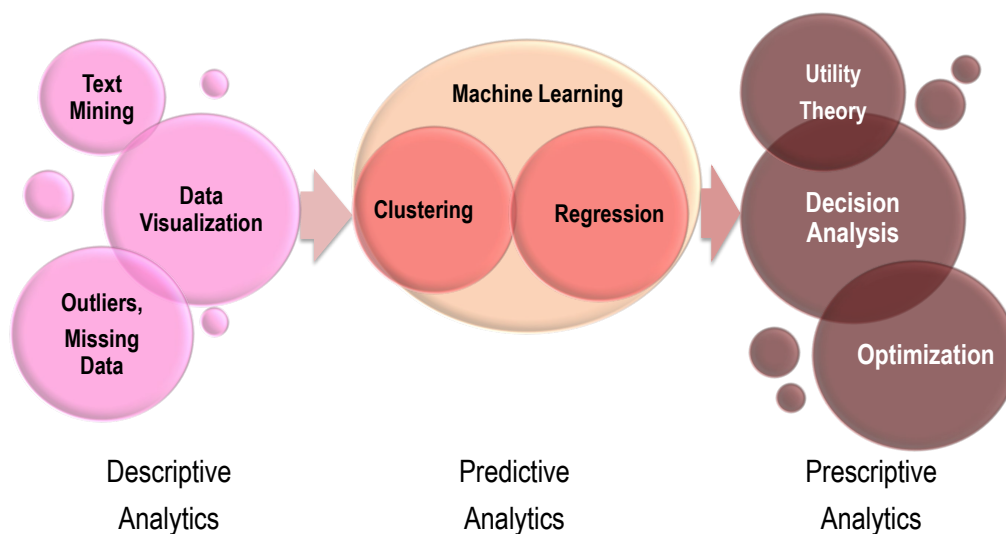
Lecture 5: Introduction to Machine Learning & Clustering

Emily Zhu, PhD.

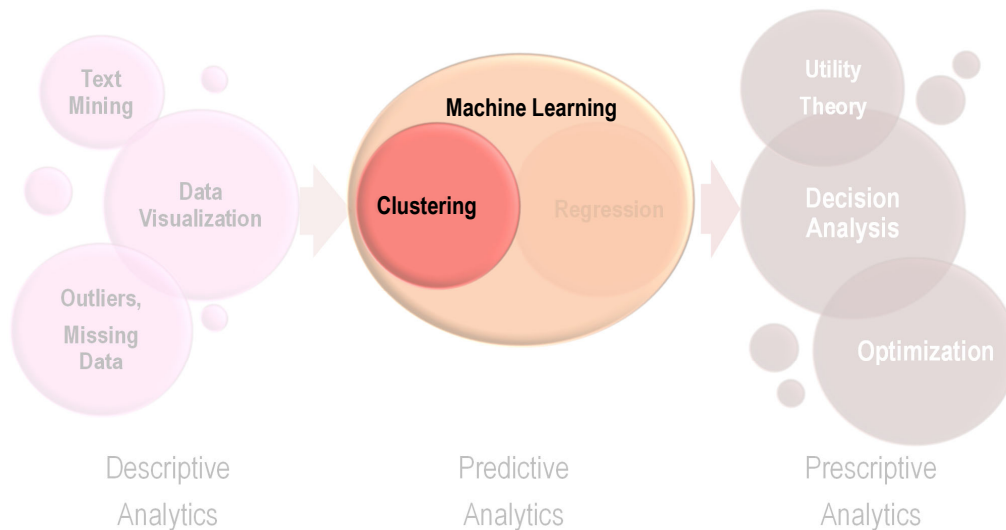
© 2023 Texas State University

Our Problem and Roadmap

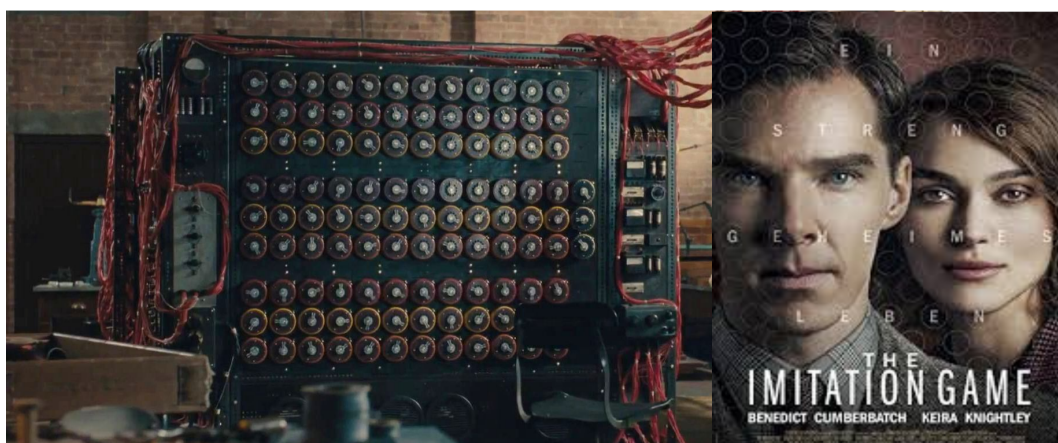
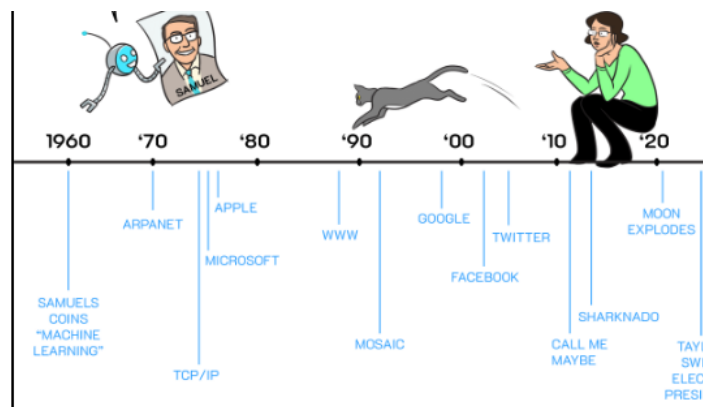
An investment property in Austin: **Airbnb** versus **Rental**?



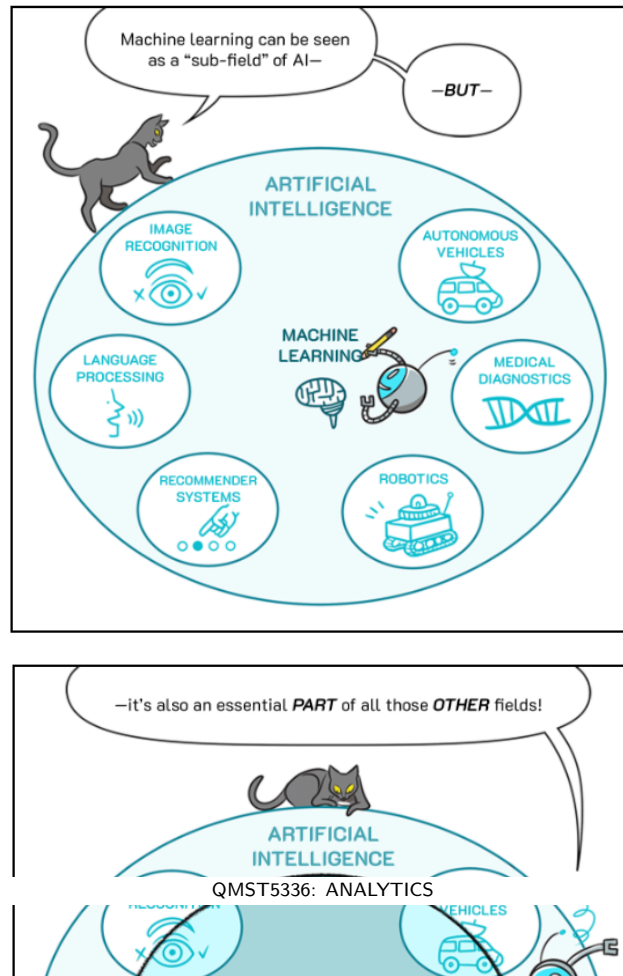
Roadmap: Where We Are



Machine Learning: A Bit History



Artificial Intelligence & Machine Learning



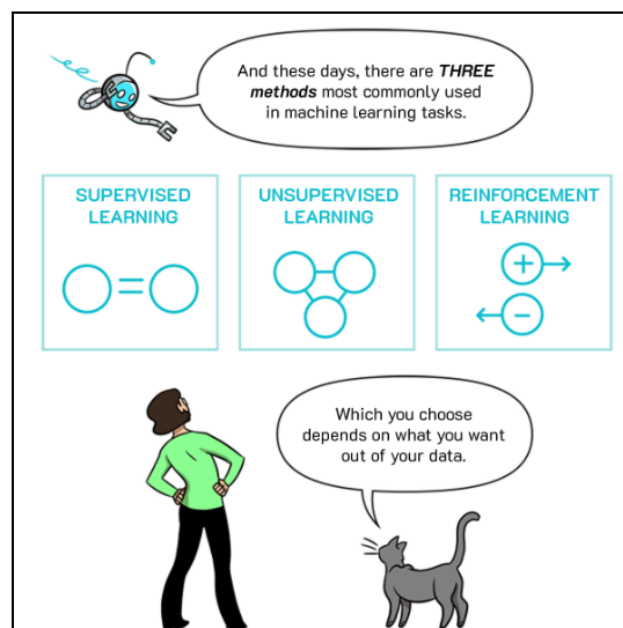
Emily Zhu, PhD.

QMST5336: ANALYTICS

© 2023 Texas State University

5 / 36

Mainstreams Of Machine Learning



Source: Machine Learning. An Online Comic From Google AI
<https://cloud.google.com/products/ai/ml-comic-1/>

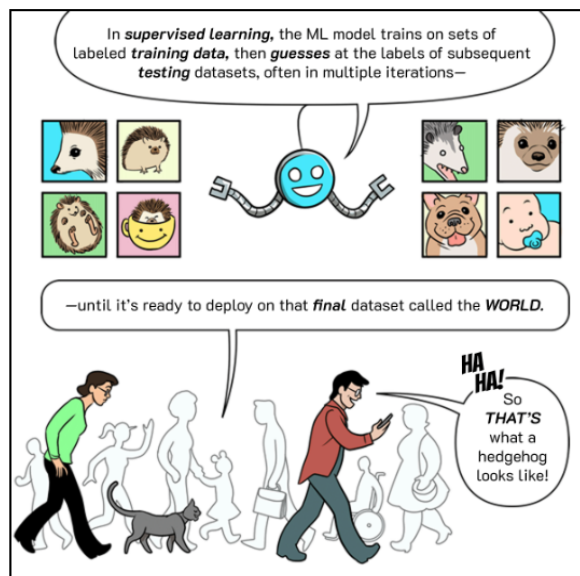
Emily Zhu, PhD.

QMST5336: ANALYTICS

© 2023 Texas State University

6 / 36

Supervised Learning



Source: Machine Learning. An Online Comic From Google AI
<https://cloud.google.com/products/ai/ml-comic-1/>

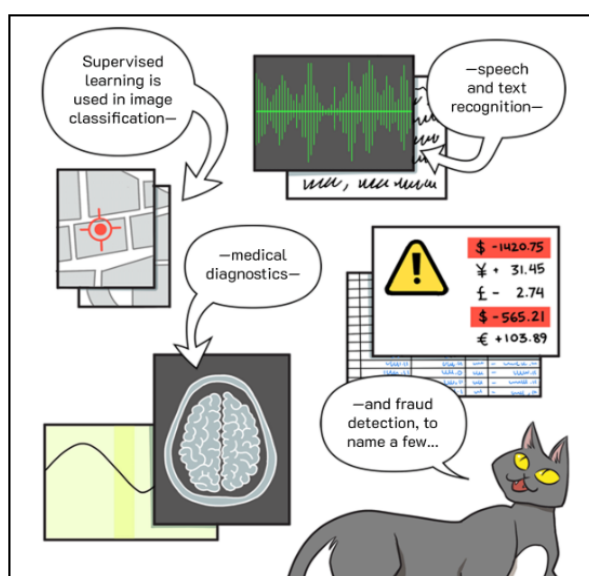
Emily Zhu, PhD.

QMST5336: ANALYTICS

© 2023 Texas State University

7 / 36

Supervised Learning: Applications



- Logistic regression (Predicting spam email)
- Classification of images (Computer vision)
- Rate prediction (restaurant, movies, music)

Source: Machine Learning. An Online Comic From Google AI
<https://cloud.google.com/products/ai/ml-comic-1/>

Emily Zhu, PhD.

QMST5336: ANALYTICS

© 2023 Texas State University

8 / 36

Supervised Learning: Natural Language Process

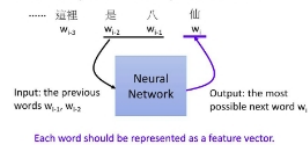
- Sentiment Analysis
- Text summarization (financial statement)
- Machine Translation
- Chatbot (Cool application here)



Sentiment analysis

Predicting the next word

- Given a sequence of words, predict the next word



Predicting next words



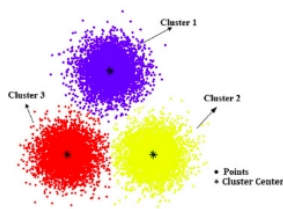
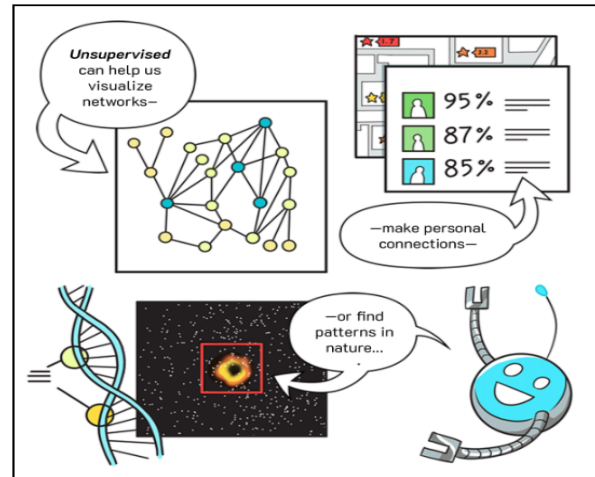
Chatbot

Unsupervised Learning

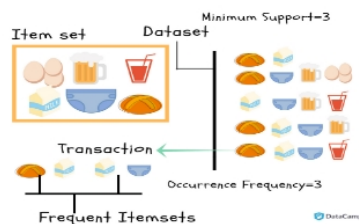


Unsupervised Learning: Applications

- Clustering
- Find association rules
- Generating realistic images



Clustering



Association rules



Generating realistic images

Emily Zhu, PhD.

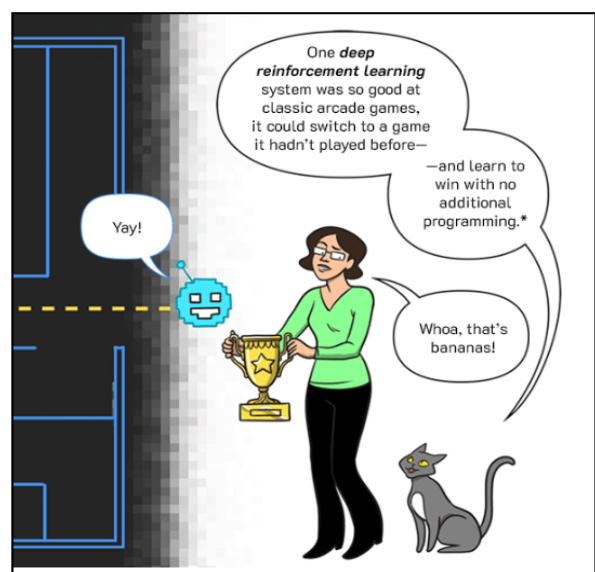
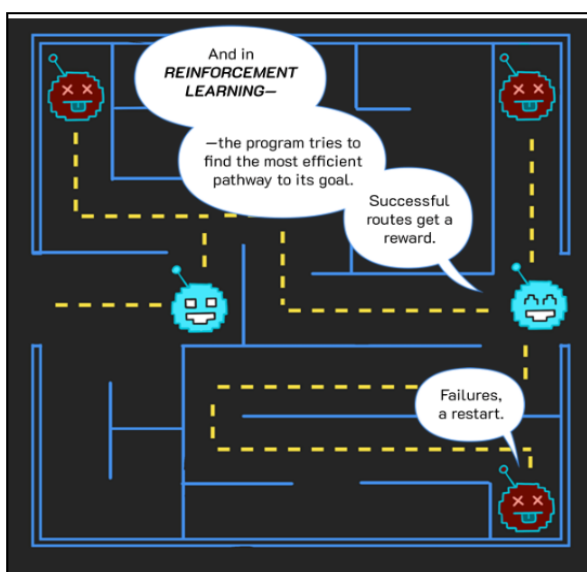
QMST5336: ANALYTICS

© 2023 Texas State University

11 / 36

<https://cloud.google.com/products/ai/ml-comic-1/>

Reinforcement Learning



Source: Machine Learning. An Online Comic From Google AI

<https://cloud.google.com/products/ai/ml-comic-1/>

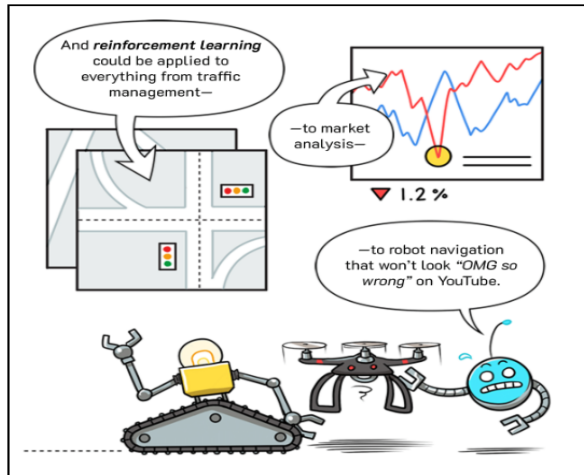
Emily Zhu, PhD.

QMST5336: ANALYTICS

© 2023 Texas State University

12 / 36

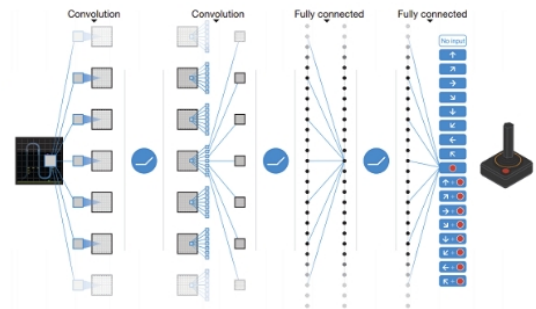
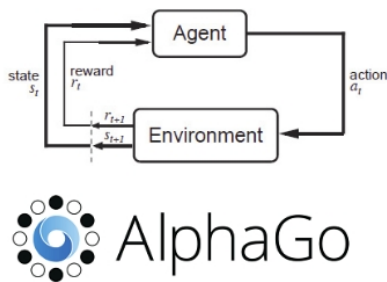
Reinforcement Learning



- Decision making/planning problems
- Take the best decision **in the long TERM**
- ex: investments banking, video games, everyday life decisions (going industry or keep studying?)



State



Emily Zhu, PhD.

QMST5336: ANALYTICS

© 2023 Texas State University

13 / 36

Source: Machine Learning. An Online Comic From Google AI
<https://cloud.google.com/products/ai/ml-comic-1/>

Deep Mind

2015

Alpha Go versus Lee Sedol



2019

Alpha Star versus Team Liquids



Figure: Demo Starcraft II (01:40:00)

<https://deepmind.com/blog/alphastar-mastering-real-time-strategy-game-starcraft-ii/>

[//deepmind.com/blog/alphastar-mastering-real-time-strategy-game-starcraft-ii/](https://deepmind.com/blog/alphastar-mastering-real-time-strategy-game-starcraft-ii/)

Emily Zhu, PhD.

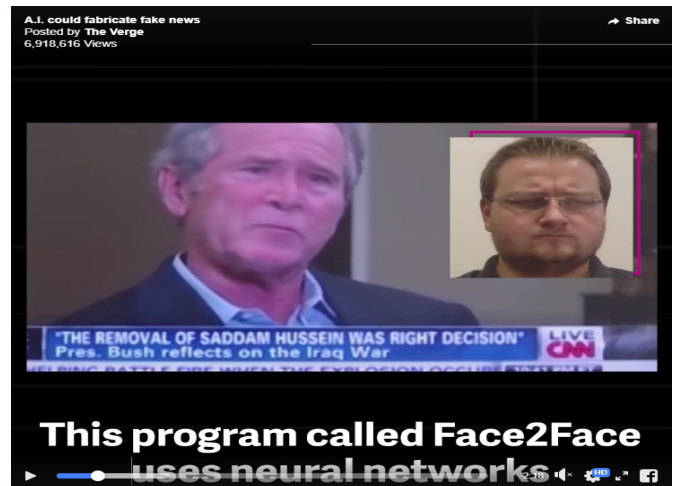
QMST5336: ANALYTICS

© 2023 Texas State University

14 / 36

Concerns/Ethical issues

- Privacy issues: Alexa's system (Amazon)
- Microsoft bot shut down in China for ethical issues
- Faking Obama (00:20)



https:

[//www.cbsnews.com/news/amazon-workers-are-listening-to-what-you-tell-alexa/](https://www.cbsnews.com/news/amazon-workers-are-listening-to-what-you-tell-alexa/)

<https://www.bbc.com/news/world-asia-china-40815024>

<https://www.theverge.com/tldr/2018/4/17/17247334/>

[ai-fake-news-video-barack-obama-jordan-peelee-buzzfeed](https://www.theverge.com/tldr/2018/4/17/17247334/ai-fake-news-video-barack-obama-jordan-peelee-buzzfeed)

Emily Zhu, PhD.

QMST5336: ANALYTICS

© 2023 Texas State University

15 / 36

Clustering: Measuring Similarity

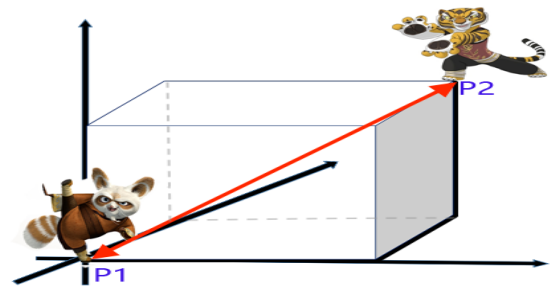
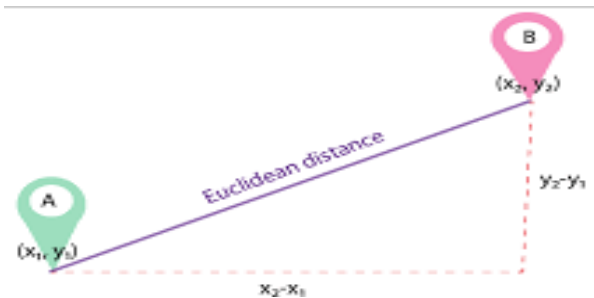
The goal of clustering is to segment observations into similar groups based on the observed variables.

Measurements of Similarity or Dissimilarity:

- maximize the similarity between observations.
- minimize the distance between observations in a cluster.

Distance: Measuring Dissimilarity

When observations include **quantitative/numerical variables**, **Euclidean distance** is the most common method to measure **dissimilarity** between observations.



Euclidean distance is highly influenced by the scale on which variables are measured.

Solution: Standardization

```
1 from sklearn import preprocessing
2 standardized_data = preprocessing.scale(raw_data)
```

Jaccard's Coefficient: Measuring similarity

When observations have **ONLY categorical variables**, **Jaccard's coefficient** is a better way to measure **similarity** between observations.

$$\frac{\text{\# of variables with matching True for observations } u \& v}{\text{total \# of variables} - \text{\# of variables with matching False for observations } u \& v}$$

u:    

v:     

u

v

						
1	1	1	1	0	0	0
1	0	1	0	1	1	1

$$\text{Jaccard's Coefficient} = \frac{2}{7}$$

Similarity Between Clusters

The similarity between two clusters can be defined by:

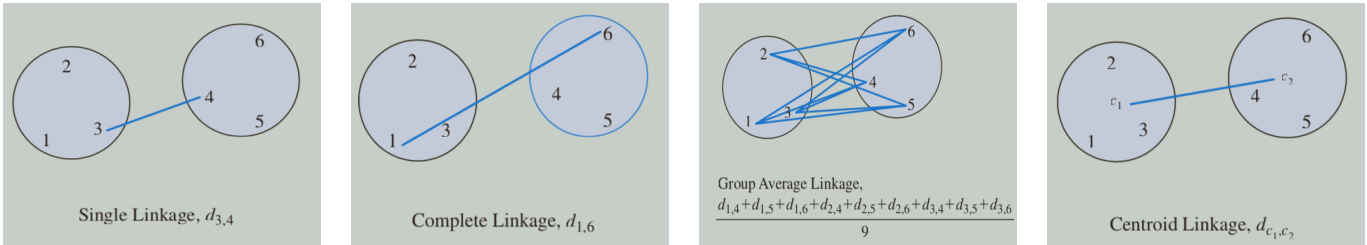
single linkage the pair of observations (one from each cluster) that are the most similar;

complete linkage the pair of observations (one from each cluster) that are the most different;

group average(median) linkage the average(median) similarity computed over *all* pairs of observations between the two clusters;

centroid linkage the distance of cluster centroid, which is calculated by the average value for each variable across all observations in a cluster; that is, a centroid is the average observation of a cluster.

ward's method the dissimilarity of the observations within the resulting single cluster increases as little as possible; that is, minimizing the variance of the clusters being merged.



Emily Zhu, PhD.

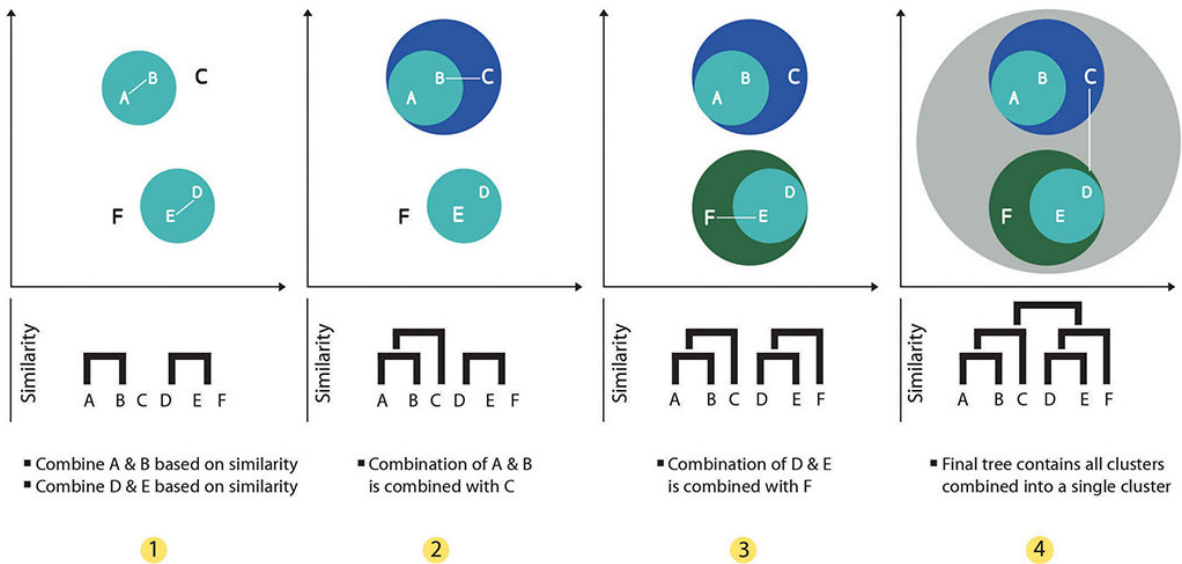
QMST5336: ANALYTICS

© 2023 Texas State University

19 / 36

Hierarchical Clustering

A bottom-up approach that starts with each observation in its own cluster and then iteratively combines the two clusters that are the most similar into a single cluster. Each iteration corresponds to an increased level of aggregation by decreasing the number of distinct cluster.



This approach can be very sensitive to outliers.

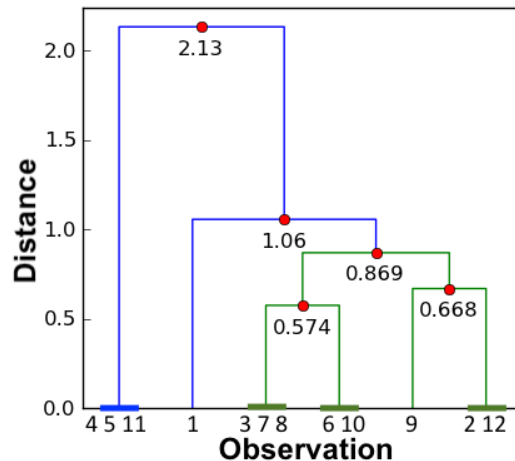
Emily Zhu, PhD.

QMST5336: ANALYTICS

© 2023 Texas State University

20 / 36

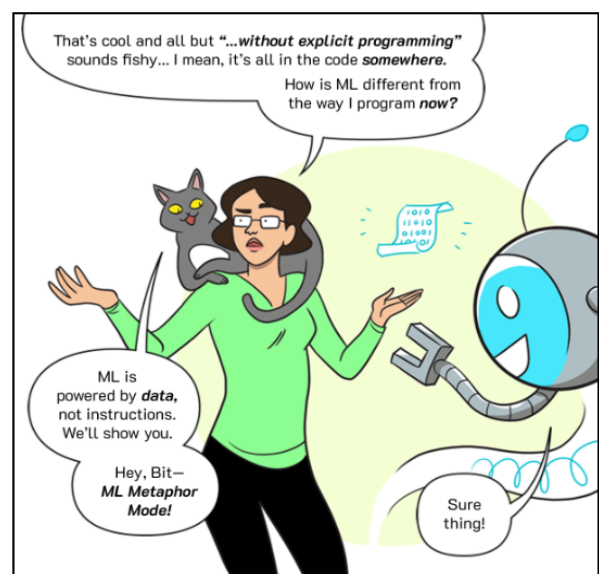
Dendrogram: Visualize Hierarchical Clustering



A dendrogram is a chart depicting the set of nested clusters resulting at each step of aggregation.

- Each horizontal line represents a merger of two (or more) clusters.
- The vertical line represents the increase of the dissimilarity (distance) resulting from a merger.
- A cluster's durability (or strength) can be measured by the difference between the distance value at which a cluster is originally formed and the distance value at which it is merged with another cluster.

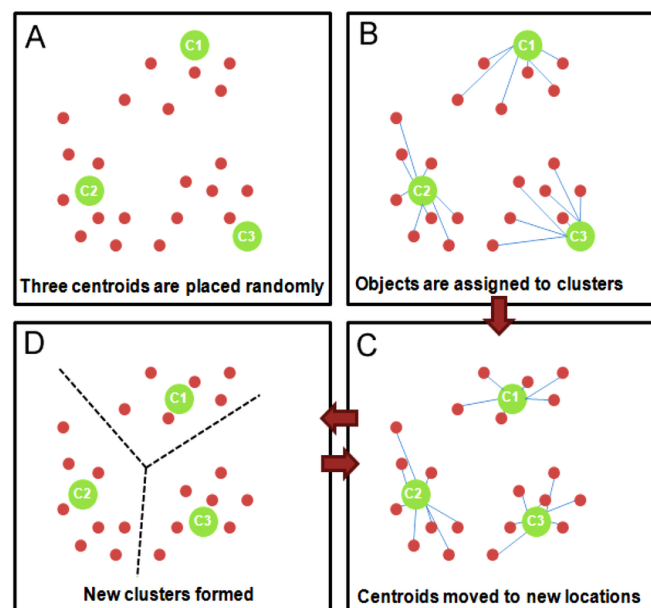
Clustering in Python



Hierarchical Clustering in Python

```
1 # import hierarchical clustering libraries
2 import scipy.cluster.hierarchy as sch
3 from sklearn.cluster import AgglomerativeClustering
4 # create dendrogram with data df_0
5 dendrogram = sch.dendrogram(sch.linkage(df_km, method='ward'))
6 # create clusters
7 hc = AgglomerativeClustering(n_clusters=4, affinity = 'euclidean',
8                               linkage = 'ward')
9 # save clusters for chart
10 y_hc = hc.fit_predict(df_1)
11 # Here you can use different data for prediction
```

K-Means Clustering



The k -means clustering algorithm can be repeated for several values of k . The algorithm repeats the following process until no change of observations in the cluster or a specified maximum number of iterations is reached.

- calculate cluster centroid;
- assign observations to cluster with nearest centroid;

KMeans in Python

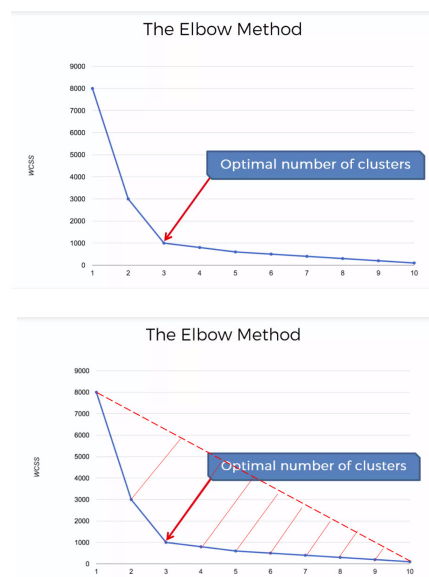
```
1 from sklearn.cluster import KMeans #must use sklearn.cluster
2 kmeans = KMeans(n_clusters = 4)
3 # fit kmeans object to data df_1
4 kmeans.fit(df_1)
5 # print location of clusters learned by kmeans object
6 print(kmeans.cluster_centers_)
7 # save new clusters for chart
8 y_km = kmeans.fit_predict(df_1)
9 # Here you can predict using different data
```

This method is generally not appropriate for categorical data where an “average” is not meaningful.

Fix k for K-Means

Within-Cluster-Sum-of-Squares (WCSS) refers to sum of distances of observations from their cluster centroids.

- WCSS decreases with the value of k ;
- Given k , K-Means minimizes WCSS, equivalently maximizes the distance between clusters.



Finding the Number of Clusters in Python

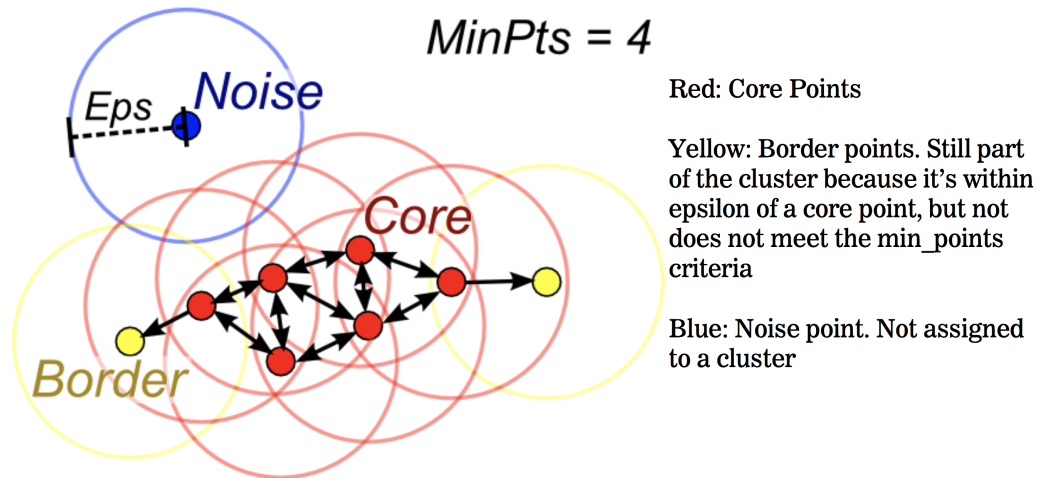
```
1 from sklearn.cluster import KMeans
2 import matplotlib.pyplot as plt
3 wcss = []
4 for i in range(1, 11):
5     kmeans = KMeans(n_clusters = i, init = 'k-means++',
6                     random_state
7                     = 42)
8     kmeans.fit(df_1)
9     wcss.append(kmeans.inertia_)
10 plt.plot(range(1, 11), wcss)
11 plt.title('K-Means')
12 plt.xlabel('Number of clusters')
13 plt.ylabel('WCSS')
14 plt.show()
```

Hierarchical V.S. K-means Clustering

Hierarchical Clustering	K-Means
Suitable when we have a small data set (e.g., fewer than 500 observations) and want to easily examine solutions with increasing numbers of clusters.	Suitable when you know how many clusters you want and you have a larger data set (e.g., more than 500 observations).
Convenient method if you want to observe how clusters are nested.	Partitions the observations, which is appropriate if trying to summarize the data with k “average” observations that describe the data with the minimum amount of error.

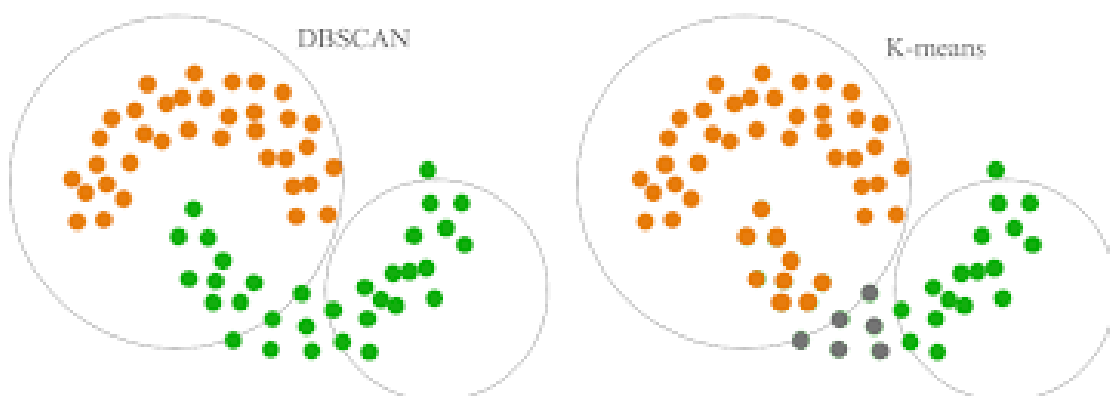
Bonus: DBSCAN Clustering

Density-based spatial clustering of applications with noise (DBSCAN)



```
1 from sklearn.cluster import DBSCAN
2 db = DBSCAN(eps = 0.02)
3 db.fit(df_1)
4 y_km = db.fit_predict(df_2)
```

KMeans V.S DBSCAN



Case: How Are the Properties Located?

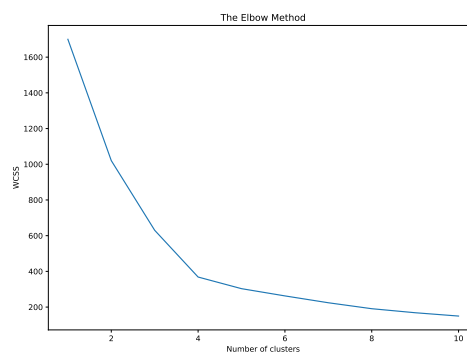
The data

```
1 df_loc = df_listing(['latitude', 'longitude'])
2 df_1 = df_loc.sample(800)
3 # A Random Sample of 800 Observations
```

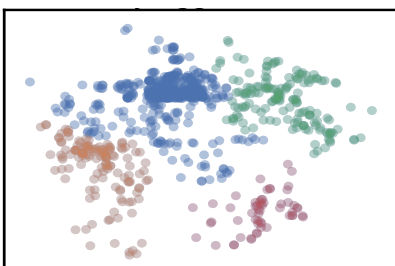
Preprocess: Standardize the scales

```
1 from sklearn import preprocessing
2 df_2 = preprocessing.scale(df_1)
```

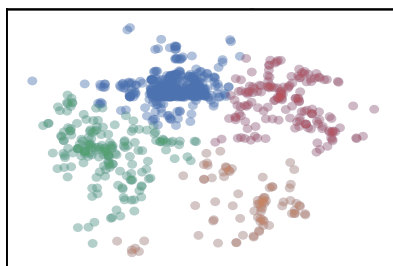
Case: Clustering Result



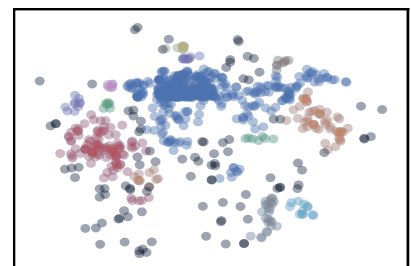
Hierarchical Cluster



K-Means



DBSCAN



Evaluate Clustering Results

The fact the lack of “truth”.

- perform a visual check if possible using common sense;
- major outliers regarding cluster cardinality (no. of examples per cluster);
- anomalies in cluster magnitude (the sum of distances from all examples to the centroid of the cluster);
- cross check with multiple methods;
- performance of downstream system.

Hierarchical Cluster



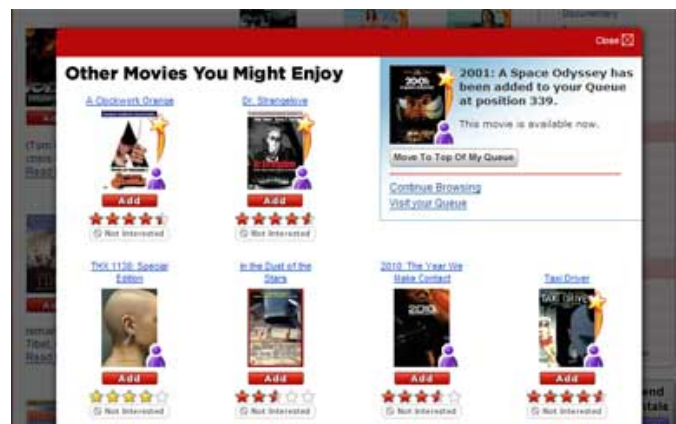
K-Means



DBSCAN



Applications of Clustering



Frequently Bought Together



Price for all three: **\$94.23**

[Add all three to Cart](#) [Add all three to Wish List](#)

[Show availability and shipping details](#)

- ✓ **This item:** 1500 Thread Count Queen 4pc Bed Sheet Set Egyptian Quality Deep Pocket White by Choice Linen **\$25.75**
- ✓ Chezmoi Collection White Goose Down Alternative Comforter, Full/Queen with Corner Tab by Chezmoi Collection **\$44.49**
- ✓ Dream Supreme Plus Gel Fiber-Filled Pillows, Standard (Set of 2) by Dream Supreme Plus **\$23.99**

Customers Who Bought This Item Also Bought

Page 1 of 11



Takeaway: Put Alikes Together

Great minds think alike



BQBA
MADE

Before Next Lecture

■ Quiz 1:

- Covers Lectures 1 to 4, inclusively.
 - 10 multiple choice questions.
 - 30 minutes.
 - Open book.
 - September 22-28, inclusively
- Assignment 1 wrapper (group assignment) due September 22nd.

